

Topics: CSS Box Model  
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Topics:

1. Box Model
2. Padding and Margin
3. Visualization of Box Model
4. CSS Positions
5. Colors

Recap:

1. CSS - Cascading Style-sheets

Style sheets - because we add in styles to our HTML structure

Cascading - because CSS can be applied from multiple locations.

Precedence Rules for CSS Application:

1. Location - Inline > Internal > External
2. Specificity - ID > Class > Elements
3. Order of appearance - Later in the file is applied
4. !important - adds a flag for higher specificity

Inclusion of CSS

- <link> <style> <style>

Inline - `<tag style="property: value;">`

Internal - `<head>`

`<style> CSS_STYLES </style>`  
`</head>`

External - using link tag in the head tag of HTML page

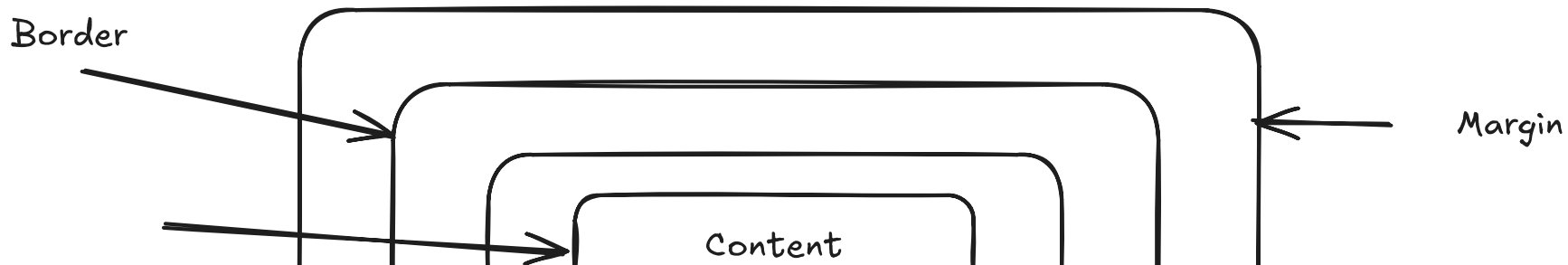
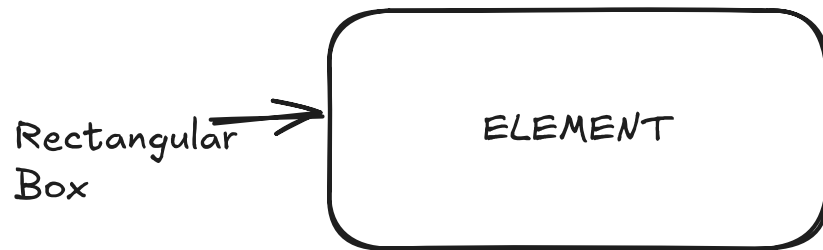
`<link rel="stylesheet" href="FILE_PATH">`

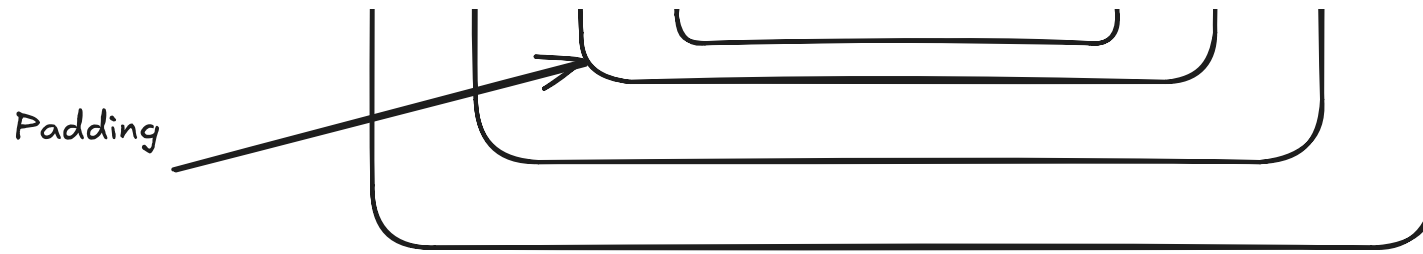
Pseudo Classes: the classes which defines rule for a specific state of the element

eg. - anchor tag having HOVER, ACTIVE, VISITED, LINK

button - HOVER, ACTIVE

## Box Model





Padding - It will control the positioning of the content within the box

Margin - It will control the positioning of the content on the page.

Content - It can be any image, paragraph, text, container

Border - it wraps the content and the padding - treat border like a frame.

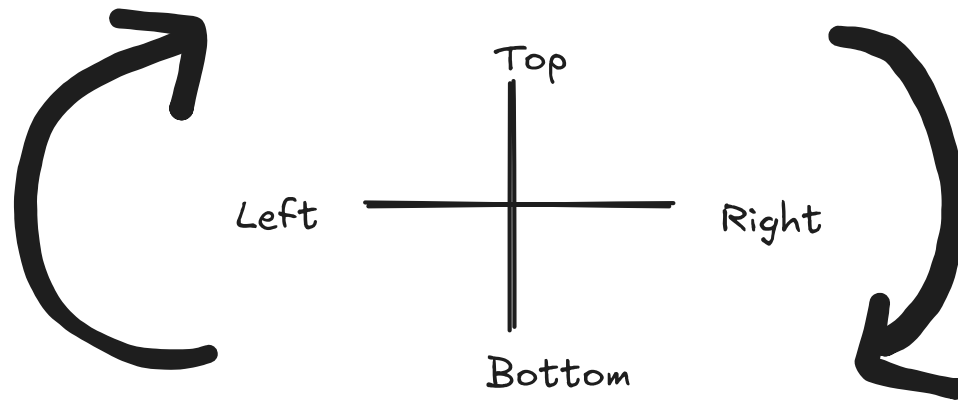
When to use margin and when to use padding?

1. We should use padding when we have to add space inside the box or we have to add space around the content without affecting its outer position
2. We should use margin to create space between elements - like separating the elements away or close to each other

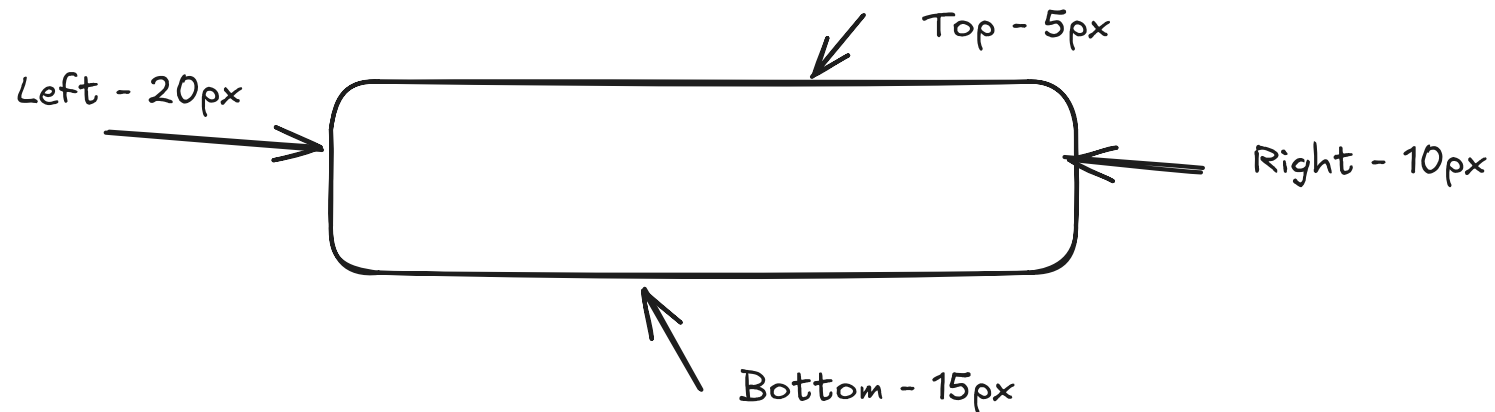
padding: 5px

margin: 5px

it means we are adding the same value on the all sides of the box.



padding: TOP RIGHT BOTTOM LEFT;  
padding: 5px 10px 15px 20px



margin: TOP RIGHT BOTTOM LEFT (t r b l)  
margin: 5px 10px 15px 20px

property: value1 value2

margin: 5px 10px

padding: 5px 10px

first value = top and bottom

second value = right and left

shorthand property:

margin: 5px; -> it will apply to all the sides

padding: 5px; -> add padding of 5px to all the sides

If we use 3 values instead of 4, the last value is treated as the same as of the right side

top - 5px

right - 10px

bottom - 15px

left - missing (it will assume the same value as of right which is 10px)

We can add margin or padding separately using the specific properties:

padding-left: 5px;

padding-bottom: 15px;

padding-right: 25px;

padding-top: 35px;

```
margin-top: 5px;  
margin-bottom: 20px;  
margin-right: 10px;  
margin-left: 15px;
```

element.style:

Adding styles to this space can help identify the spacing or look and feel of the content and tag placement.

Can also be useful for debugging for any spacing issues.

display: inline

it can have its very own height and width. It will continue from where the last element left off.

display: block

it will always begin with a new line. and has control over its own height and width

display: inline-block

it will combine the properties of both inline and block. meaning it will continue from where the previous element left off and can control the height width of the element

Using em and rem as margin or padding:

Yes, it is possible.

But to apply this unit, we have to define the font-size in the parent or root element.

using auto:

margin: 0 auto

It will horizontally center the element given that a fixed width is provided

CSS Positioning:

1. static
2. relative
3. absolute
4. fixed

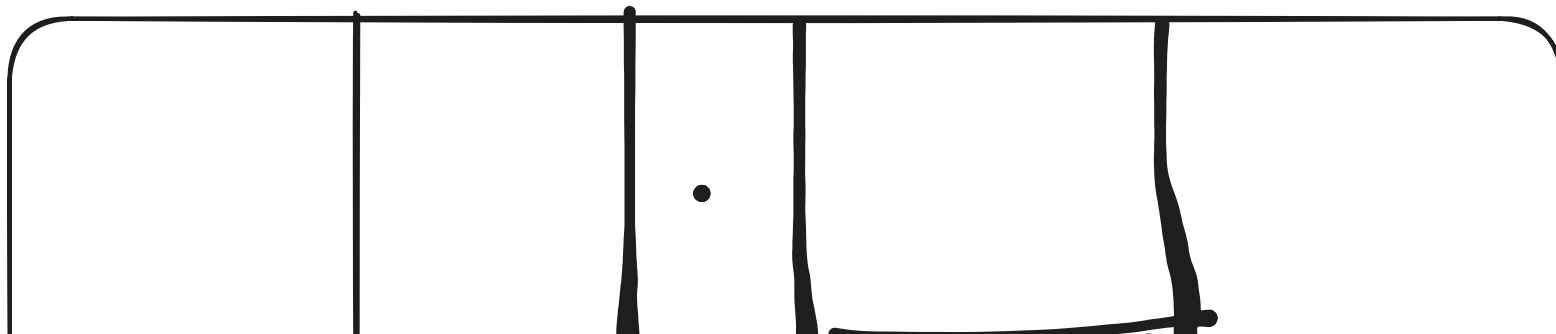
Default is set to static.

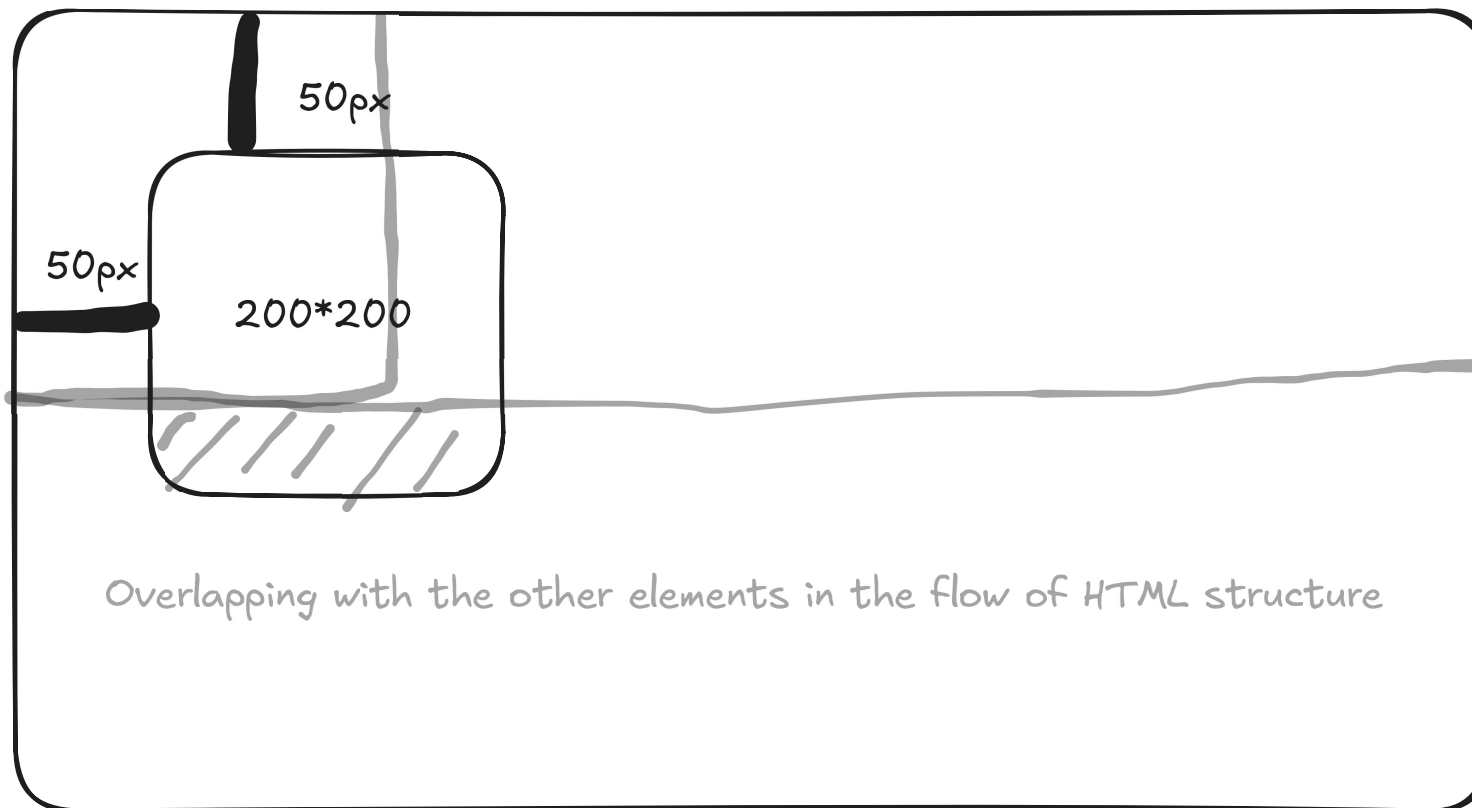
Static - it will just follow the natural flow of the document

relative - the element stays in the flow, but we can move it in different directions  
as in top, bottom, left or right and the space is reserved for the element

absolute - it takes the element out of the flow and positions it relative to nearest parent  
with relative or absolute position

fixed - sticks to the viewport so that it stays at the place when we scroll.





CSS Colors - Solid, RGBA, HEX

1. solid - blue, red, orange, yellow

2. Hex - #000000 to #ffffff

3. RGB - combination of red, green and blue colors. which are the primary colors  
`rgb(red_value, green_value, blue_value)`

4. RGBA - it is similar to the RGB, with a touch of transparency to it. It is alpha  
`rgba(255, 0, 0, 1).`

Alpha resides in the range of 0 to 1

Red, green and blue value are in the range 0 to 255

5. HSL: it is easier to change visually, hue, saturation and lightness

`hsl(0, 100%, 50%)`

hue - zero degree to 359 degree. if we move to 360 it will be considered as 0

saturation - zero to hundred percent

lightness - zero to hundred zero - dark or black

hundred - white

Color Gradients:

Gradient allows us to blend two or more colors.

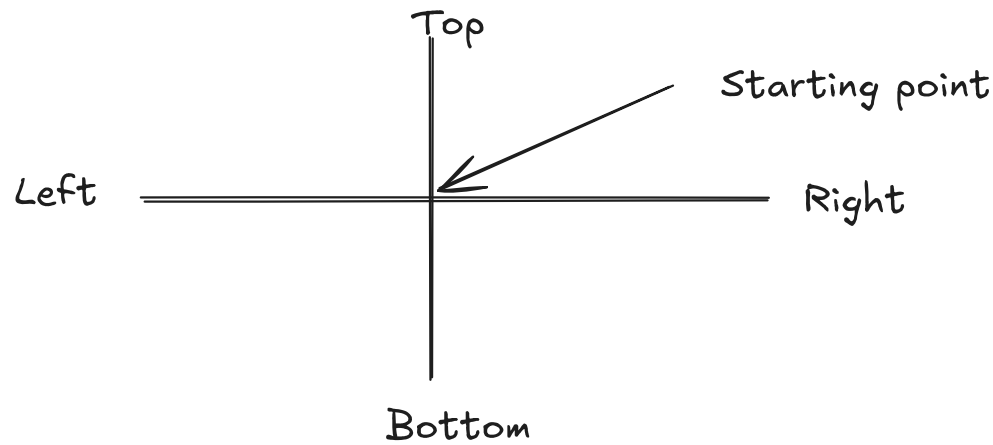
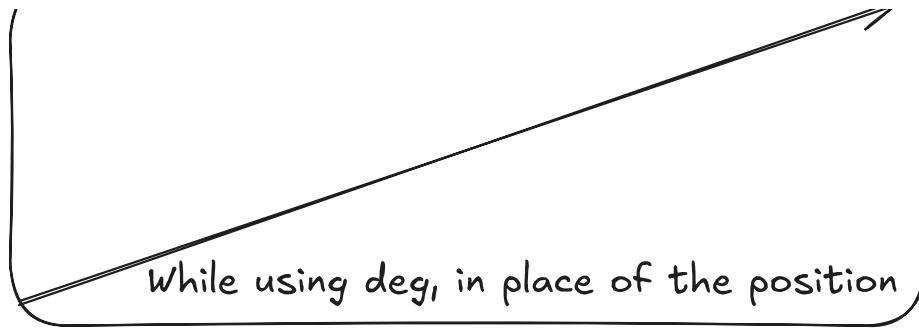
`linear-gradient(add the properties)`

`linear-gradient(to right, yellow, red)`

`linear-gradient(position of the gradient, color1 percentage, color2 percentage, ....)`

where percentage is the place where the color should be completely transitioned to the mentioned color





- 0deg -> from starting point to the top (to top)
- 90deg -> towards right (to right)
- 180deg -> towards the bottom (to bottom)
- 270deg -> towards the left (to left)

Task:

1. Creating a gradient with background image
2. Add some gradients to the portfolio page

3. Understand the box model and its changes.